

NoteTrade

Team Name: NoteTraders

NoteTrade is a web platform that allows students to buy and sell study resources such as notes, textbooks, and other learning materials.

The platform aims to make it easier for students to find affordable study resources, while also allowing sellers to clear items they no longer need. Instead of relying on informal chats or small friend groups, NoteTrade provides a more organised space where students can browse listings and manage their own items.

Motivation

Students often exchange notes and textbooks through informal channels such as messaging apps or friend groups. However, this limits the number of people who can view, buy, or sell these resources.

NoteTrade aims to provide a more open and organised platform where students can easily browse available study resources, while sellers can list items they no longer need.

User Stories

- As a student seller, I want to create an account and log in securely, so that I can manage my listings and account details.
- As a student seller, I want to create listings for my used notes or textbooks, so that other students can view and buy them.
- As a student seller, I want to edit or delete my listings, so that I can update item details or remove items that are no longer available.
- As a student buyer, I want to browse available listings, so that I can find second-hand notes or textbooks that I need.
- As a student buyer, I want to view listing details such as title, description, price, and seller information, so that I can decide whether I am interested in the item.
- As a user, I want to update my account particulars, so that my personal information stays accurate.
- As a user, I want to receive clear error messages when something goes wrong, so that I know what to correct.

- As a user, I want usernames to be unique, so that each account can be clearly identified.

Features Implemented

1. Authentication (core)

[Proposed]

Allows users to create an account and log in using a username and password. To enhance security, the platform will also support two-factor authentication (2FA), where users verify their identity using a one-time password (OTP) sent to their registered email address or phone number. This ensures that only authorised users can access their accounts, manage listings, and perform transactions on the platform.

[Current progress]

We have created a MongoDB collection hosted on MongoDB Atlas for accounts that can record the username, password, email and phone number for each account

When creating an account: the platform will check if the username exists already, if so, it will prevent the new user from creating an account with a duplicate username Passwords will need to be at least 8 characters, consisting of 1 uppercase alphabet, 1 lowercase alphabet, 1 number and 1 special character The new account will have a unique account id once created Users can upload a profile photo for their account

Users will be able to update their account details such as username, password, email and phone number. Users are also able to login to their account if they provide the correct username and password.

[Future works]

By Milestone 2: Hash the passwords stored in the database to increase security
Keep track of the amount of money that belongs to an account (related to the transaction feature)

By Milestone 3: Enable users to delete their account Enable users to save their profile picture

2. Listing of items (core)

[Proposed]

Allows sellers to create listings for their study materials by placing them under the appropriate category and providing key details such as the title, description, and price. This helps organise resources clearly and makes it easier for buyers to browse and understand what is being offered.

[Current progress]

We have created a collection on MongoDB that keeps track of the listing title, description, price, academic level, subject. Listings can be created, displayed, updated and deleted.

[Future Works]

By Milestone 2: Link the listing to the sellers Enable pictures of the listing to be saved

3. Transaction of items (core)

[Proposed]

Allows users to top up money into their account balance and use this balance to purchase items on the platform. Sellers will be able to receive payments through the platform, while users can also withdraw their balance easily. This provides a simple and convenient way to facilitate transactions between buyers and sellers.

[Current progress]

None

[Future works]

By Milestone 2: Implement this feature so that when a buyer presses “buy”, the balance remaining in the buyers account decreases, while the sellers’ account balance increases

4. Search and Filter (core)

[Proposed]

Allows users to search for listings using a search bar and narrow down results through filters such as academic level, module code, subject, price, and category. This improves the browsing experience by helping users quickly find study materials that are relevant to their needs.

[Current Progress]

None

[Future Works]

By Milestone 2: Implement a search bar where users can search for keywords in the listing title or description Implement a filter function whereby users can filter listing via price range, academic level, and subject

By Milestone 3: Allow users to sort listings by alphabetical order or by price (increasing and decreasing)

5. Chat (extension)

[Proposed]

Allows interested buyers to message sellers directly to ask for more information about the listed item or to negotiate the price. This enables smoother communication between users and helps buyers make more informed decisions before purchasing.

[Current progress]

None

[Future Works]

By Milestone 3: Implement the chat feature between two users

6. Favourites (extension)

[Proposed]

Allows buyers to save or like listings that they are interested in so they can revisit them later. This gives users a convenient way to keep track of items without having to search for them again.

[Current Progress]

None

[Future works]

By Milestone 3: Implement the favourites for each account by adding another field under the accounts collection that keeps track of the favourite listings' id

Tech Stack

Frontend: React, TypeScript

Backend: Go, Gin

Database: MongoDB

Version Control: GitHub

API Routes Used

Method	Route	Description
POST	<code>/signup</code>	Creates a new user account
POST	<code>/login</code>	Logs a user into their account
POST	<code>/updateUser</code>	Updates user particulars
POST	<code>/createListing</code>	Creates a new listing
POST	<code>/updateListing</code>	Updates an existing listing
POST	<code>/deleteListing</code>	Deletes a listing
GET	<code>/listings</code>	Retrieves all listings
GET	<code>/users</code>	Retrieves all users

Run Locally

Clone the project

```
git clone https://github.com/TYS2/NoteTraders
```

Go to the project directory

```
cd NoteTraders
```

Running the Backend

Go to the backend folder

```
cd backend
```

Install dependencies

```
go mod tidy
```

Run the backend server

```
go run main.go
```

Running the Frontend

Go to the frontend folder

```
cd frontend
```

Install dependencies

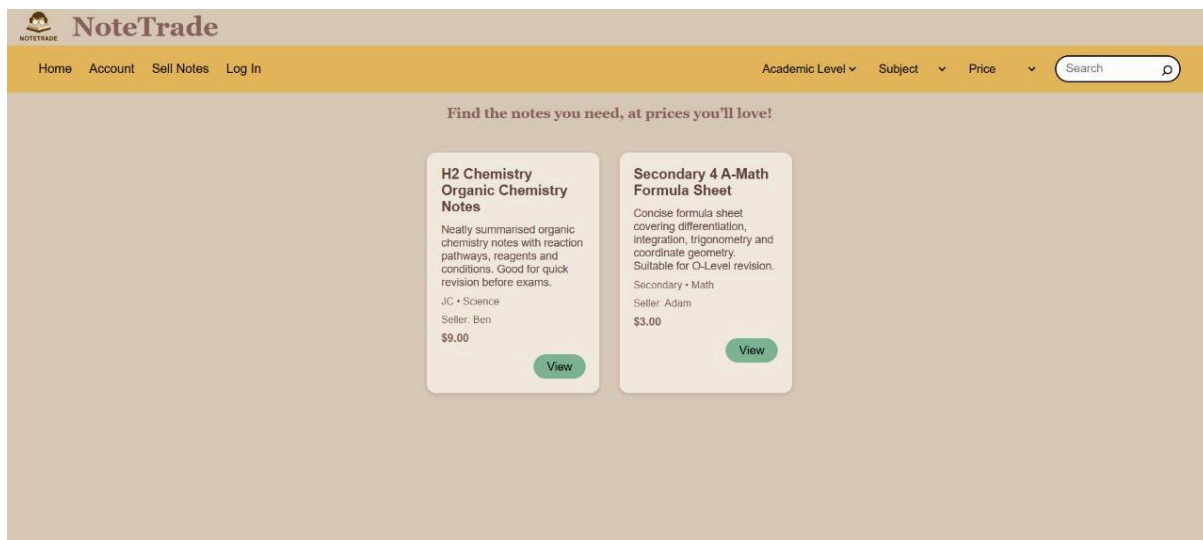
```
npm install
```

Start the server

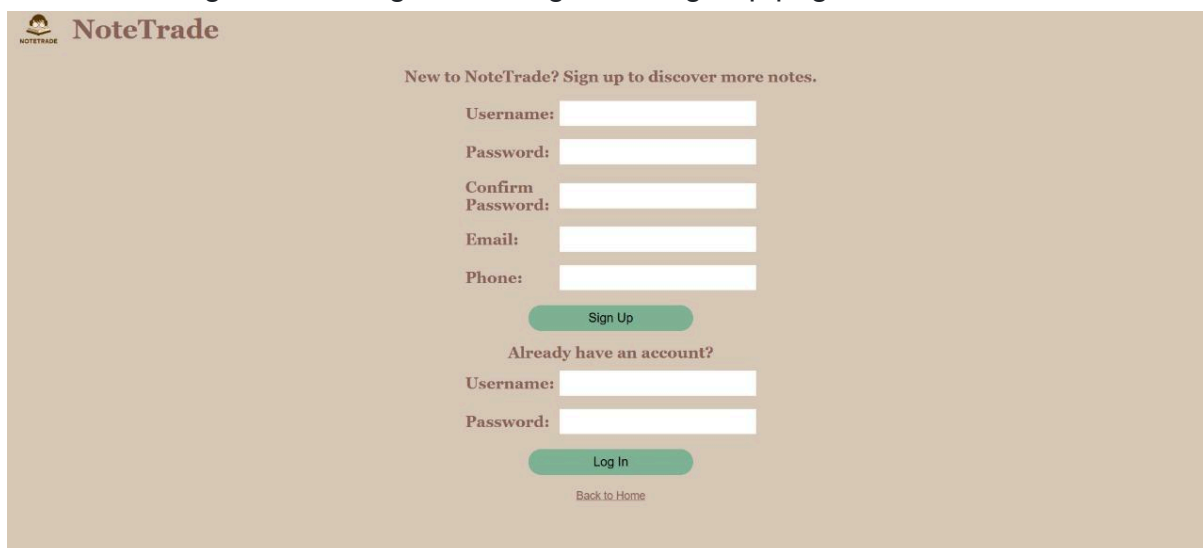
```
npm run dev
```

How to use

1. Open the website and view the listings on the homepage.



2. Click on the login button to go to the login and sign-up page.



3. Sign up for a new account or log in with an existing account.



NoteTrade

New to NoteTrade? Sign up to discover more notes.

Username:

Password:

Confirm Password:

Email:

Phone:

[Sign Up](#)

Already have an account?

Username:

Password:

[Log In](#)

[Back to Home](#)



NoteTrade

New to NoteTrade? Sign up to discover more notes.

Username:

Password:

Confirm Password:

Email:

Phone:

[Sign Up](#)

Already have an account?

Username:

Password:

[Log In](#)

[Back to Home](#)

4. After logging in, continue browsing the listings on the homepage.



NoteTrade

Home Account Sell Notes Log Out Academic Level Subject Price Search

Welcome back, Adam!

Find the notes you need, at prices you'll love!

H2 Chemistry Organic Chemistry Notes

Neatly summarised organic chemistry notes with reaction pathways, reagents and conditions. Good for quick revision before exams.

JC • Science

Seller: Ben

\$9.00

[View](#)

Secondary 4 A-Math Formula Sheet

Concise formula sheet covering differentiation, integration, trigonometry and coordinate geometry. Suitable for O-Level revision.

Secondary • Math

Seller: Adam

\$3.00

[View](#)

5. Go to the account page to view your account details and the listings you have created.

The screenshot shows the 'My Account' page of the NoteTrade application. The header includes the NoteTrade logo, navigation links (Home, Account, Sell Notes, Log Out), and filters for Academic Level, Subject, and Price, along with a search bar. The 'My Account' section displays the following details: Account ID: 1, Username: Adam, Email: adam@gmail.com, and Phone Number: 91234567. There is an 'Edit Particulars' button and a profile picture placeholder with an 'Upload Picture' button. The 'Account Balance' is shown as '-', with buttons for 'Top Up', 'Withdraw', and 'View Transaction History'. The 'My listings' section shows a listing titled 'Secondary 4 A-Math Formula Sheet' with a description, category 'Secondary • Math', and price '\$3.00'. It has 'Edit', 'View', and 'Delete' buttons. An 'Add' button is also present. The 'My Chats' section is visible at the bottom.

6. Edit your account details if needed. If the username already exists, an error message will be shown.

This screenshot shows the 'My Account' page with the 'Edit Particulars' form active. The 'Username' field is highlighted and contains the text 'Adam'. Other fields (Email: adam@gmail.com, Phone Number: 91234567) and buttons ('Save', 'Cancel', 'Upload Picture', 'Top Up', 'Withdraw', 'View Transaction History') are visible. The 'My listings' and 'My Chats' sections remain the same as in the previous screenshot.

This screenshot shows the 'My Account' page after an attempt to update the username. A red error message is displayed at the top: 'Failed to update particulars: Username already exists'. The 'Edit Particulars' form is still active, but the 'Username' field now contains 'Ben'. The 'My listings' and 'My Chats' sections are unchanged.

7. Create, view, edit, or delete your own listings.

 **NoteTrade**

HomeAccountSell NotesLog Out

Academic Level▼Subject▼Price▼

Create a Listing

Title

Description

Price

Academic Level

Choose one▼

Subject

Choose one▼

Post Listing

 **NoteTrade**

HomeAccountSell NotesLog Out

Academic Level▼Subject▼Price▼

Secondary 4 A-Math Formula Sheet

Concise formula sheet covering differentiation, integration, trigonometry and coordinate geometry. Suitable for O-Level revision.

Secondary • Math
Seller: Adam

\$3.00

BuyChat with seller

[Back to Home](#)

 **NoteTrade**

HomeAccountSell NotesLog Out

Academic Level▼Subject▼Price▼

My Account

Account ID: 1
Username: Adam
Email: adam@gmail.com
Phone Number: 91234567
Edit Particulars

Account Balance: -
Top UpWithdrawView Transaction History


Upload Picture

My listings

Secondary 4 A-Math Formula Sheet

Concise formula sheet covering differentiation, integration, trigonometry and coordinate geometry. Suitable for O-Level revision.

3

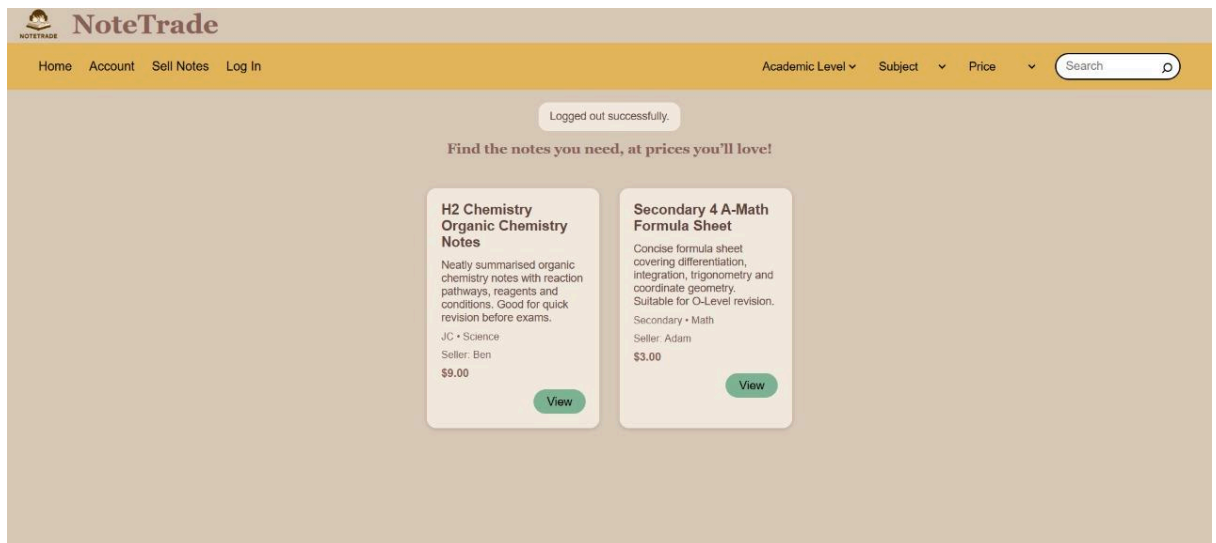
Secondary▼

Math▼

SaveCancel

My Chats

8. Log out of your account when you are done.



Team Members

- Teng Ywee See
- Xue XinYu